



Menopause: Clinical features and diagnosis

AUTHOR: [Robert F Casper, MD](#)

SECTION EDITOR: [Robert L Barbieri, MD](#)

DEPUTY EDITOR: [Jean E Mulder, MD](#)

All topics are updated as new evidence becomes available and our [peer review process](#) is complete.

Literature review current through: **Aug 2026**.

This topic last updated: **Sep 04, 2025**.

INTRODUCTION

Natural menopause is defined as the permanent cessation of menstrual periods, determined retrospectively after a woman has experienced 12 months of amenorrhea without any other obvious pathologic or physiologic cause. It occurs at a median age of 51.4 years and is a reflection of complete, or near complete, ovarian follicular depletion, with resulting hypoestrogenemia and high follicle-stimulating hormone (FSH) concentrations ([figure 1](#)).

The menopausal transition (perimenopause) occurs after the reproductive years, but before menopause, and is characterized by irregular menstrual cycles, endocrine changes, and symptoms such as hot flashes.

The terms for natural menopause that occurs between the ages of 40 and 45 years or <40 years are "early menopause" and "primary ovarian insufficiency," respectively. Both early menopause and primary ovarian insufficiency have been associated with an excess risk of cardiovascular disease, bone loss, and cognitive function concerns. This topic will review the clinical features and diagnosis of the menopausal transition and menopause and briefly review the criteria for primary ovarian insufficiency and early menopause. The clinical manifestations and health impact of elective bilateral oophorectomy at the time of hysterectomy are reviewed separately. In addition, the physiology and epidemiology of menopause, postmenopausal hormone therapy, and primary ovarian insufficiency are reviewed separately:

- (See "[Elective oophorectomy or ovarian conservation at the time of hysterectomy](#)".)
- (See "[Ovarian development and failure \(menopause\) in normal women](#)".)

- (See ["Menopausal hormone therapy: Benefits and risks"](#).)
 - (See ["Pathogenesis and causes of spontaneous primary ovarian insufficiency \(premature ovarian failure\)"](#).)
-

STAGES OF REPRODUCTIVE AGING

STRAW staging system — The Stages of Reproductive Aging Workshop (STRAW) staging system was developed based upon data from multiple longitudinal cohort studies [1]. It is considered the gold standard for characterizing reproductive aging from the reproductive years through menopause and includes criteria for the reproductive years, menopausal transition, final menstrual period (FMP), and postmenopause based upon bleeding patterns, endocrine findings, and symptoms ([figure 2](#)). The menopausal transition and postmenopause are further subdivided into "early" and "late" stages.

Although the STRAW system has been used primarily for women's health research, it may also be helpful in the clinical setting for patients and clinicians to assess fertility potential, contraceptive needs, and potential need for hormone therapy. We find the bleeding and symptom criteria of STRAW to be useful when counseling patients about what to anticipate in the coming years ([figure 2](#)).

Of note, the STRAW staging criteria are not considered to represent diagnostic criteria for the menopausal transition or menopause, primarily because they include endocrine data (follicle-stimulating hormone [FSH], inhibin B, anti-müllerian hormone [AMH]) and pelvic ultrasound (antral follicle count [AFC]) as supportive criteria when determining reproductive stage. All four criteria have been used to assess ovarian reserve in the setting of assisted reproductive technologies, but none have been validated for use in the evaluation of menopausal status. (See ["Female infertility: Evaluation"](#), [section on 'Assessment of ovarian reserve'](#) and ["In vitro fertilization: Procedure"](#), [section on 'Assessment of ovarian reserve'](#).)

Virtually all women experience the menstrual irregularity and hormonal fluctuations prior to clinical menopause; up to 80 percent develop hot flashes (the most common menopausal symptom), but only 20 to 30 percent seek medical attention for them ([algorithm 1](#)). (See ["Hot flashes"](#) below.)

Much of the available information about the endocrine and clinical manifestations of the menopausal transition comes from longitudinal cohort studies of midlife women [2-14], the largest of which, the Study of Women's Health Across the Nation (SWAN), has followed a multiethnic, community-based cohort of over 3000 women ages 42 to 52 years for 15 years [2,7-9,11,12,15-24]. Based upon data from these cohort studies, a staging system was developed that is now considered to be the gold standard for characterizing reproductive aging from the reproductive years through menopause ([figure 2](#)).

Late reproductive years — In the late reproductive years before the onset of the menopausal transition, serum inhibin B (a granulosa cell protein marker reflecting oocyte number) begins to decrease [25], allowing the rise in serum follicle-stimulating hormone (FSH) while estradiol levels are preserved. Luteal phase progesterone levels decrease as fertility potential begins to decline ([figure 3](#)). Inhibin B is not typically measured to assess reproductive stage.

Menstrual cycles are ovulatory, but the follicular phase (the first half of the menstrual cycle before ovulation occurs) begins to shorten (eg, 10 versus 14 days) ([figure 4](#)) [26]. Women who are having difficulty conceiving often seek advice about their menopausal status during this stage. Although there is variability in age at any given stage of reproductive aging, women are typically in their 40s when cycles begin to shorten. (See "[Female infertility: Evaluation](#)", [section on 'Assessment of ovarian reserve'](#).)

Menopausal transition (perimenopause) — The menopausal transition begins, on average, four years before the FMP and includes a number of physiologic changes that may affect a woman's quality of life. It is characterized by irregular menstrual cycles and marked hormonal fluctuations, often accompanied by hot flashes, sleep disturbances, and mood symptoms [1,27-31] (see '[Clinical features](#)' below). In addition, changes in lipids and bone loss begin to occur, both of which have implications for long-term health. The term "perimenopause" includes the menopausal transition years as well as the first year after the FMP.

As the process of ovarian follicular depletion continues in midlife ([figure 1](#)), women eventually experience a change in intermenstrual interval. This change in bleeding pattern, which is accompanied by hormonal fluctuations and a variety of symptoms, is referred to as the menopausal transition, or perimenopause, and occurs on average at age 47 years [32].

- **The early transition** – The early transition is defined by a change of ≥ 7 days in the intermenstrual interval. Normal intermenstrual interval during the reproductive years is 25 to 35 days; during the menopausal transition, this may increase to 40 to 50 days or there may be a shortening of the intermenstrual interval, similar to that seen in the late reproductive years. Early follicular phase FSH levels, while variable, are typically high ([figure 5](#)). This initial stage of the menopausal transition is referred to as the "early transition" in the Stages of Reproductive Aging Workshop (STRAW) staging system, which is described below ([figure 2](#)). (See '[STRAW staging system](#)' above.)
- **The late transition** – After the initial lengthening of the intermenstrual interval, women progress to more dramatic menstrual cycle changes with skipped cycles, episodes of amenorrhea, and an increasing frequency of anovulatory cycles ([figure 5](#)). This stage is referred to as the "late transition" in the STRAW staging system and typically lasts for one to three years before the FMP ([figure 2](#)) [1] (see '[STRAW staging system](#)' above). Of note,

not all women follow a "typical" bleeding pattern. Some will have episodes of amenorrhea interspersed with short ovulatory cycles that resemble those of the late reproductive stage.

- **Endocrine findings** – The more irregular cycles are often accompanied by dramatic fluctuations in serum FSH and estradiol concentrations ([figure 5](#)). Consequently, a random serum sample may demonstrate high FSH and low estradiol concentrations consistent with menopause, but soon thereafter, FSH and estradiol may return to the normal premenopausal range ([figure 5](#)) [33]. One study reported that a random serum FSH >25 international units/L is characteristic of the late menopausal transition [29], but measurements of serum FSH during the late menopausal transition are not routinely recommended, due to their variability and lack of predictability for menopausal onset or symptomatology. (See '[Evaluation](#)' below.)

Other endocrine changes across the menopausal transition include a progressive decrease in serum inhibin B, as well as a decrease in anti-müllerian hormone (AMH), another product of the ovary's granulosa cells. In addition, ovarian antral follicle count (AFC), defined as the remaining complement of follicles measuring 2 to 10 mm in diameter on transvaginal ultrasound, declines steadily across the reproductive years though postmenopause.

AMH, AFC, and cycle day 3 FSH and estradiol have all been used to assess ovarian reserve in the setting of assisted reproductive techniques, but none are validated for the evaluation of menopausal status [1]. Inhibin B is typically measured during a diagnostic evaluation for ovarian tumors, but not for evaluation of menopause. (See "[Female infertility: Evaluation](#)", [section on 'Assessment of ovarian reserve'](#).)

- **Menstrual bleeding** – In general, the transition is characterized by a gradual decrease in menstrual bleeding [15]. However, some women experience heavy or prolonged bleeding, which has always been assumed to be due to anovulatory cycles and prolonged exposure of the endometrium to unopposed estrogen. However, in one report, the heaviest bleeding occurred in women in the late transition during ovulatory cycles, which were more likely than anovulatory cycles to be associated with high serum estradiol concentrations [34] and, as noted above, a relative progesterone deficiency. Women with obesity and uterine fibroids are also more likely to experience heavy bleeding [15]. The evaluation and management of heavy (>80 mL) and prolonged bleeding (>7 days) is reviewed separately. (See "[Abnormal uterine bleeding in nonpregnant reproductive-age patients: Management](#)".)

Menopause — After the years of menstrual irregularity, women eventually experience permanent cessation of menses or clinical menopause. The FMP is defined retrospectively after 12 months of amenorrhea. The first two to six years after the FMP are referred to as "early postmenopause" in the STRAW staging system and "late postmenopause" thereafter. (See '[STRAW staging system](#)' above.)

Although the median age at natural menopause is 51.4 years, the timing of menopause is affected by a number of factors, including genetics, smoking, and reproductive history, which are reviewed separately. (See "[Ovarian development and failure \(menopause\) in normal women](#)", [section on 'Epidemiology'](#).)

- **Serum FSH** – The increase in serum FSH becomes sustained near the FMP, then increases over several years to levels in the 70 to 100 international units/L range, followed by a decline with increasing age [35,36].
- **Predicting the FMP** – Being able to estimate the timing of the FMP has greater importance than simply allowing women to know when they will reach menopause, as accelerated bone loss and an increase in cardiovascular risk factors occur in the year leading up to the FMP. (See '[Long-term consequences of estrogen deficiency](#)' below.)

Methods for predicting the FMP (based upon menstrual cycle changes, age, and/or serum FSH and estradiol concentrations) have been proposed, but are not used routinely [7,37-39]. A serum AMH-based predictor appears to be superior to FSH-based predictors for estimating when a woman will undergo their FMP [40]. This predictor only applies to women over age 42 and requires the use of a specific, highly sensitive AMH assay.

CLINICAL FEATURES

Hot flashes — The hallmark symptom of the menopausal transition/perimenopause and early postmenopausal years is the hot flash. Women may experience a number of other symptoms including sleep disturbances and new-onset depression [41]. For other symptoms such as joint pain and memory loss, the association with menopause is less clear.

- **Epidemiology** – The most common symptom during the menopausal transition and menopause are hot flashes (also referred to as vasomotor symptoms or hot flushes), which occur in up to 80 percent of women [16,41-43]. However, only approximately 20 to 30 percent of women seek medical attention for treatment [17,44,45]. Some women first develop hot flashes that cluster during the nadir of estradiol secretion that occurs in the late luteal and early follicular phases during their late reproductive years, but symptoms are typically mild and do not require treatment. Symptoms become far more common during the menopausal transition, with a frequency of approximately 40 percent in the early transition, increasing to 60 to 80 percent in the late menopausal transition and early postmenopausal periods ([figure 2](#) and [figure 6](#)) [17,18,32,42,46]. When hot flashes occur at night, women typically describe them as "night sweats."
- **Symptoms** – Hot flashes typically begin as the sudden sensation of heat centered on the upper chest and face that rapidly becomes generalized. The sensation of heat lasts from two to four minutes, is often associated with profuse perspiration and occasionally

palpitations, and is sometimes followed by chills and shivering, and a feeling of anxiety. Hot flashes usually occur several times per day, although the range may be from only one or two each day to as many as one per hour during the day and night. Hot flashes are particularly common at night. (See "[Menopausal hot flashes](#)".)

- **Geographical variability** – There is considerable geographic variability in the prevalence of menopausal hot flashes. Among American and Japanese females, the prevalence of moderate to severe hot flashes is 35 and 15 percent, respectively [47]. In a survey of menopausal females in five European countries, the overall prevalence of moderate to severe hot flashes was slightly higher than in the United States (40 versus 35 percent), ranging from a low of 31 percent in France to a high of 52 percent in Italy [48]. The reasons for the variation are unclear; smoking rates were similar in each country and body mass indexes were not provided.
- **Duration** – It had been thought that vasomotor symptoms diminish and stop within a few years of onset in most women. However, vasomotor symptoms can persist for as long as 20 years past the FMP and may be variable between racial/ethnic groups [49]. A more detailed discussion of the pathophysiology, risk factors for, duration, and treatment of hot flashes is found elsewhere. (See "[Menopausal hot flashes](#)".)
- **Impact on function** – Women with seven or more daily moderate to severe vasomotor symptoms report interference with sleep (94 percent), concentration (84 percent), mood (85 percent), energy (77 percent), and sexual activity (61 percent) [50]. The more severe the vasomotor symptoms, the more daily activities are impacted, leading women to seek treatment to improve quality of life.
- **Impact on work** – Hot flashes and other menopausal symptoms have a negative impact on work outcomes (absences, decreased hours, layoffs, firings, and job changes specifically due to menopause symptoms). In a survey study of 4440 menopausal women with moderate symptom scores based upon the Menopause Rating Scale (MRS), 480 (11 percent) reported missing work in the preceding year (average of three days missed) [51]. Women with more severe symptoms (in the highest quartile of MRS scores) were approximately 15 times more likely to have an adverse work outcome than those in the lowest quartile. Based on workdays missed due to menopause symptoms, the authors estimated an annual \$1.8 billion direct loss to the United States economy.

Depression — There is a significant increased risk of new-onset depression in women during the menopausal transition compared with their premenopausal years [19,52-58]. The risk then decreases in the early postmenopause. In a within-woman, eight-year longitudinal study to determine risk factors for depressive disorders, a diagnosis of depression was 2.5 times more likely to occur in the menopausal transition compared with when the woman was premenopausal (odds ratio [OR] 2.50; 95% CI 1.25-5.02) [56]. This association is most marked

for women with a prior history of depression or a mood problem. The role of estrogen in the treatment of depression in perimenopausal women is discussed separately. (See ["Treatment of menopausal symptoms with hormone therapy", section on 'Mood lability/depression'.](#))

Sleep disturbance — A distressing feature of hot flashes is that they are more common at night than during the day and are associated with arousal from sleep. However, peri- and postmenopausal women experience sleep disturbances even in the absence of hot flashes [59]. The estimated prevalence of difficulty sleeping based upon two longitudinal cohort studies was 32 to 40 percent in the early menopausal transition, increasing to 38 to 46 percent in the late transition [43,60].

Anxiety and depression symptoms, which are common during the menopausal transition, may contribute to sleep disturbances; in one study, they were predictive of subjective sleep disturbances [61]. In addition, perimenopausal women with hot flashes are more likely to be depressed [62,63]. (See ["Depression"](#) above.)

Primary sleep disorders are also common in this population. In a report of 102 women ages 44 to 56 years who reported sleep disturbances, 54 (53 percent) had sleep apnea, restless legs syndrome, or both [61].

Thus, in peri- or postmenopausal women who report sleep disturbances, treating the vasomotor symptoms may decrease sleep disturbances, but this may not resolve all sleep problems, as there are many other things that can disturb sleep, such as primary sleep disorders, anxiety, and depression [61]. (See ["Risk factors, comorbidities, and consequences of insomnia in adults"](#).)

Cognitive changes — While biologic and epidemiologic evidence suggests that estrogen is important for cognitive function in women, the consequences of hormonal changes during the menopausal transition, estrogen deficiency after menopause, and the impact of estrogen therapy remain uncertain. While many women experience cognitive issues during the menopausal transition, these symptoms are typically transient. (See ["Estrogen and cognitive function"](#).)

During the menopausal transition, some women describe symptoms such as forgetfulness, difficulties with word retrieval, and "brain fog" [64]. However, in a cross-sectional study of over 200 men and women ages 45 to 55, women in their late reproductive or perimenopausal years (determined by menstrual status and serum estradiol and FSH concentrations) outperformed age-matched men in detailed memory tasks [65]. These apparent sex differences were attenuated in the postmenopausal years. Higher serum estradiol concentrations in women, a reflection of residual ovarian activity, were associated with better performance. Thus, some evidence suggests that in early midlife, women appear to

outperform men on memory tasks, but with the onset of menopause and decline in serum estradiol, any memory advantage diminishes. (See ["Estrogen and cognitive function"](#).)

A decline in cognitive function was not observed in the SWAN study, but increases in anxiety and depression had independent, unfavorable effects on cognitive performance [20]. (See ["Stages of reproductive aging"](#) above and ["Estrogen and cognitive function"](#).)

Genitourinary syndrome of menopause — Genitourinary syndrome of menopause (GSM), formerly referred to as vulvovaginal atrophy, affects 50 percent of menopausal women and is defined as a collection of symptoms and signs caused by hypoestrogenic and hypoandrogenic changes to the labia, clitoris, vagina, urethra, and bladder that occur in menopausal women. In general, these symptoms develop later than other menopausal symptoms like hot flashes and may affect sexual functioning, emotional well-being, body image, and interpersonal relations [66]. GSM was initially thought to be related to estrogen deficiency, but both estrogen and androgen deficiency may play a role [67]. GSM leads to a decrease in blood flow to the vagina and vulva resulting in decreased vaginal lubrication and sexual dysfunction in menopausal women. Symptoms related to GSM are exquisitely responsive to vaginal estrogen therapy. Vaginal androgen therapy (DHEA) may possibly complement estrogen action [66,68]. (See ["Genitourinary syndrome of menopause: Clinical manifestations and diagnosis of vulvovaginal symptoms"](#), section on 'Epidemiology' and ["Overview of sexual dysfunction in females: Epidemiology, risk factors, and evaluation"](#) and ["Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy"](#), section on 'Vaginal estrogen therapy'.)

Other

- **Joint aches and pain** – Joint aches and pain are a commonly reported symptom among women at midlife [69-72], with a reported prevalence as high as 50 to 60 percent in cross-sectional studies [69,71]. While women who are obese or depressed are more likely to experience joint pain, there also appears to be an association with menopausal status, with peri- and postmenopausal women experiencing more joint pain than premenopausal women [73]. It is unclear if the pain is related to estrogen deficiency or a rheumatologic disorder, but in the Women's Health Initiative (WHI), women with joint pain or stiffness at baseline were more likely to get relief with either combined estrogen-progestin therapy [74] or unopposed estrogen [75] than with placebo.
- **Breast pain** – Breast pain and tenderness are common in the early menopausal transition but begin to diminish in the late menopausal transition [43]. This is probably due to the fluctuations in serum estradiol concentrations. (See ["Breast pain"](#).)
- **Menstrual migraines** – Menstrual migraines are migraine headaches that cluster around the onset of each menstrual period. In many women, these headaches worsen in frequency

and intensity during the menopausal transition [76]. (See ["Estrogen-associated migraine headache, including menstrual migraine"](#).)

Long-term consequences of estrogen deficiency — Ovarian estradiol production and secretion decreases and stops altogether after menopause as a result of ovarian follicular depletion. However, the ovary continues to secrete testosterone. There are a number of long-term effects of estrogen deficiency, including osteoporosis, cardiovascular disease, and dementia. Each of these is discussed in detail separately.

- **Bone loss** – Bone loss begins during the menopausal transition. The annual rates of bone mineral density loss appear to be highest during the one year before the FMP through two years after. This issue and postmenopausal osteoporosis are discussed separately. (See ["Epidemiology and etiology of premenopausal osteoporosis"](#), [section on 'Perimenopausal bone loss'](#) and ["Overview of the management of low bone mass and osteoporosis in postmenopausal women"](#).)
- **Cardiovascular disease** – The risk of cardiovascular disease increases after menopause, thought to be at least in part due to estrogen deficiency. This may be mediated in part by changes in cardiovascular risk factors such as lipid profiles that begin to change during perimenopause. This was illustrated by longitudinal data from over 2500 subjects in the SWAN study [21]. After adjusting for subject age, there was a small increase in serum low-density lipoprotein (LDL) during the menopausal transition (a 6 percent increase in mean LDL from 116 mg/dL in the premenopausal years to 123 mg/dL in the early postmenopausal years). There was no change in serum high-density lipoprotein (HDL), but data from a later SWAN ancillary study suggested that the protective effect of HDL may decrease as women transition to menopause [22].
- **Dementia** – There is limited epidemiologic support for the hypothesis that estrogen preserves overall cognitive function in women without dementia. However, in the WHI, both unopposed estrogen and combined estrogen-progestin therapy had no global cognitive benefits in older postmenopausal women without dementia. (See ["Estrogen and cognitive function"](#), [section on 'Epidemiologic evidence'](#).)
- **Osteoarthritis** – Estrogen deficiency after menopause may contribute to the development of osteoarthritis, but data are limited. (See ["Pathogenesis of osteoarthritis"](#), [section on 'Risk factors'](#).)
- **Body composition** – In the early postmenopausal years, women who do not take estrogen therapy typically gain fat mass and lose lean mass. Some, but not all, studies, suggest that postmenopausal hormone therapy is associated with a decrease in central fat distribution. Although women typically gain weight during midlife, it does not appear to be due to

menopausal status or stage [23]. (See "[Menopausal hormone therapy: Benefits and risks](#)", section on 'Weight'.)

- **Skin changes** – The collagen content of the skin and bones is reduced by estrogen deficiency. Decreased cutaneous collagen may lead to increased aging and wrinkling of the skin. Limited data suggest collagen changes may be minimized with estrogen. (See "[Menopausal hormone therapy: Benefits and risks](#)", section on 'Skin'.)
- **Balance** – Impaired balance in postmenopausal women may be a central effect of estrogen deficiency. Problems with balance may play a role in the incidence of forearm fractures in women. The role of estrogen therapy on falls is discussed elsewhere. (See "[Menopausal hormone therapy: Benefits and risks](#)", section on 'Falls'.)

EVALUATION

In our experience, among women who present at midlife for evaluation of possible menopausal transition or menopause, many are interested in postmenopausal hormone therapy, while others simply want to know what to expect in the coming years: bleeding patterns, symptoms, or potential long-term consequences of estrogen deficiency (eg, osteoporosis, coronary heart disease, or dementia) ([algorithm 1](#)).

General approach — The evaluation depends on the patient's age (<40 years, 40 to 45 years, or over 45 years). For women of all ages, we start with an assessment of the woman's menstrual cycle history (ideally with a menstrual calendar) and a detailed history of any menopausal symptoms (hot flashes, sleep disturbances, depression, vaginal dryness). All women with symptoms of vaginal dryness, dyspareunia, or sexual dysfunction should have a pelvic examination to evaluate for vaginal atrophy.

Women over age 45 years — Although the menopausal transition begins, on average, at age 47 years [32], the age at onset of the menopausal transition is variable, and women over age 45 years who present with characteristic menopausal signs and symptoms are more likely to be in the menopausal transition than to have a new endocrine disorder. Therefore, our approach in women >45 years is as follows:

- For women who present with irregular menstrual cycles with menopausal symptoms such as hot flashes, mood changes, or sleep disturbance, we suggest no further diagnostic evaluation, as they are highly likely to be in the menopausal transition ([algorithm 1](#)).
- For women who present with amenorrhea, the possibility of pregnancy must always be considered. We obtain a serum human chorionic gonadotropin (hCG) level in sexually active women who are not using reliable contraception.

- For women with irregular menstrual cycles or amenorrhea who have additional features suggestive of hyperprolactinemia or thyroid disease (galactorrhea, goiter, tachycardia, proptosis, etc), we perform additional endocrine testing (eg, prolactin and thyroid-stimulating hormone [TSH]). (See "[Clinical manifestations and evaluation of hyperprolactinemia](#)" and "[Overview of the clinical manifestations of hyperthyroidism in adults](#)" and "[Clinical manifestations of hypothyroidism](#)".)

Although serum FSH is often measured, it is not necessary to make the diagnosis and, if normal, may be misleading. In the Study of Women's Health Across the Nation (SWAN) longitudinal cohort study described above, changes in menstrual bleeding patterns were a better predictor of menopausal stage or FMP than serum FSH concentrations [24].

Serum FSH concentrations vary widely during the menopausal transition, so a value in the normal premenopausal range does not rule out perimenopause as the cause of symptoms ([algorithm 1](#)) (see '[Menopausal transition \(perimenopause\)](#)' above). A random serum sample may demonstrate high FSH and low estradiol concentrations consistent with menopause, or normal values in the premenopausal range ([figure 5](#)) [33]. One study reported that a random serum FSH >25 international units/L is characteristic of the late menopausal transition [29], but measurements of serum FSH during the late menopausal transition are not routinely recommended, because of their variability.

Early menopause: Ages 40 to 45 years — For women between 40 and 45 years who present with irregular menstrual cycles, with or without menopausal symptoms, possible diagnoses include early menopause (if FMP occurs before age 45) or the menopausal transition. We suggest the same endocrine evaluation as for any woman with oligo/amenorrhea. This would include laboratory testing to exclude the following:

- **Pregnancy** – Serum hCG
- **Hyperprolactinemia** – Serum prolactin
- **Hyper-/hypothyroidism** – Serum TSH

Serum FSH concentrations are also typically measured but are not required to make the diagnosis of early menopause, and results are variable ([figure 5](#)) [33].

Although the presence of hot flashes with irregular menses strongly suggests the menopausal transition, we prefer to err on the side of caution and look for other possible causes of oligo/amenorrhea. (See "[Evaluation and management of secondary amenorrhea](#)", [section on 'Initial evaluation'](#).)

Primary ovarian insufficiency: Age <40 years — For women under age 40 years with irregular menses and menopausal symptoms, we suggest a complete evaluation for irregular menses. If primary ovarian insufficiency (premature ovarian failure) is confirmed, further evaluation for this disorder should be performed. (See "[Evaluation and](#)

management of secondary amenorrhea", section on 'Initial evaluation' and "Clinical manifestations and diagnosis of primary ovarian insufficiency (premature ovarian failure)".)

Atypical hot flashes — For women of any age with atypical hot flashes or night sweats, evaluation for other disorders such as carcinoid, pheochromocytoma, or underlying malignancy is indicated. This issue is discussed in detail elsewhere. (See 'Hot flashes' above.)

Heavy bleeding — Women with heavy (>80 mL) or prolonged (>7 days) bleeding should undergo the same evaluation as any premenopausal woman, eg, a pregnancy test, determine if the bleeding is ovulatory or anovulatory, rule out structural abnormalities with pelvic ultrasound, and perform an endometrial biopsy if indicated. This issue is reviewed in detail separately. (See "Abnormal uterine bleeding in nonpregnant reproductive-age patients: Management".)

DIAGNOSIS

Typical women — In women over age 45 years:

- We make the diagnosis of the menopausal transition or "perimenopause" based upon a change in intermenstrual interval with or without menopausal symptoms (hot flashes, sleep disturbance, depression, vaginal dryness or sexual dysfunction) ([algorithm 1](#) and [figure 2](#)) (see 'Menopausal transition (perimenopause)' above). A high serum follicle-stimulating hormone (FSH) concentration is **not** required to make the diagnosis.

There is no reliable method for predicting the final menstrual period (FMP) for women in the menopausal transition, although a highly sensitive AMH level (in older women) has a strong correlation with time to menopause [40]. However, women in the "late transition," as defined by the Stages of Reproductive Aging Workshop (STRAW) criteria, are likely to be closer to the FMP than women in the "early transition" ([figure 2](#)) [1].

- We diagnose menopause as 12 months of amenorrhea in the absence of other biologic or physiologic causes ([algorithm 1](#)). A high serum FSH is not required to make the diagnosis.

In women between the ages of 40 and 45 years:

- The diagnosis of the menopausal transition or early menopause is the same as that for women over 45 years, except that other causes of menstrual cycle dysfunction must first be ruled out (eg, endocrine evaluation for nonmenopausal causes of oligo/amenorrhea must be normal, including serum human chorionic gonadotropin [hCG], prolactin, and thyroid-stimulating hormone [TSH]) ([algorithm 1](#)).

For women under age 40 years:

- Women in this age group with a change in intermenstrual interval and menopausal symptoms should not be diagnosed with either the menopausal transition or menopause. They have primary ovarian insufficiency (premature ovarian failure). The biology and natural history are different and are reviewed separately. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)".)

Special situations

Women with underlying menstrual cycle disorders — The diagnosis of menopausal transition is more difficult, and the STRAW staging system does not apply to women with underlying menstrual disorders such as polycystic ovary syndrome (PCOS) or hypothalamic amenorrhea. Little information is available about menstrual cycle and endocrine changes in either disorder, but some data suggest that women with PCOS may develop more regular cycles in their later reproductive years, for reasons that are unclear [77,78]. For women with either diagnosis who develop menopausal symptoms, we suggest measuring FSH concentration for diagnostic purposes.

Women taking oral contraceptives — Oral estrogen-progestin contraceptives are considered to be safe in nonsmokers up to the age of menopause (average age 50 to 51 years) [79]. Women taking them want reassurance that they are postmenopausal before stopping. However, it is difficult to determine if menopause has occurred because these women do not develop the irregular bleeding or vasomotor symptoms that are typical of the menopausal transition. In addition, because their hypothalamic-pituitary axis is suppressed by the high dose of exogenous estrogen, measurement of the serum FSH level is unreliable. Some clinicians measure serum FSH concentration on day 7 of the pill-free interval, but we do not suggest this approach because FSH is typically still suppressed and in the premenopausal range.

We suggest stopping the pill and measuring serum FSH two to four weeks later. A level ≥ 25 international units/L indicates that the patient has likely entered the menopausal transition. However, there is no FSH value that would provide absolute reassurance that a patient is postmenopausal. We typically stop the pill by age 50 to 51 years, when the chance of conceiving is extremely low. If menopausal symptoms occur, the possibility of short-term menopausal hormone therapy for symptom relief can be discussed. (See "[Treatment of menopausal symptoms with hormone therapy](#)".)

Posthysterectomy or endometrial ablation — Menopause in women who have undergone hysterectomy or endometrial ablation cannot be determined using menstrual bleeding criteria. Therefore, supportive criteria, including assessment of menopausal symptoms and biochemical data, are needed. In this setting, we suggest measurement of FSH concentrations ([figure 5](#)). A serum FSH >25 international units/L, particularly in the setting of hot flashes, is suggestive of the late menopausal transition [29]. For a

postmenopausal woman, FSH would be considerably higher (in the 70 to 100 international units/L range) [[11](#)].

DIFFERENTIAL DIAGNOSIS

Hyperthyroidism should always be considered in the differential diagnosis, as irregular menses, sweats (although different from typical hot flashes), and mood changes are all potential clinical manifestations of hyperthyroidism [[80](#)]. (See "[Overview of the clinical manifestations of hyperthyroidism in adults](#)".)

Other etiologies for menstrual cycle changes that should be considered include pregnancy, hyperprolactinemia, and hyper- or hypothyroidism. (See "[Evaluation and management of secondary amenorrhea](#)", section on 'Initial evaluation'.)

Atypical hot flashes and night sweats may be due to other disorders, such as medications, carcinoid, pheochromocytoma, or underlying malignancy. These are discussed in detail elsewhere. (See "[Evaluation of the patient with night sweats or generalized hyperhidrosis](#)" and '[Hot flashes](#)' above.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Menopause](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Menopause \(The Basics\)](#)" and "[Patient education: Perimenopause \(The Basics\)](#)")

- Beyond the Basics topic (see "[Patient education: Menopause \(Beyond the Basics\)](#)")
-

SUMMARY AND RECOMMENDATIONS

- **Menstrual cycle and endocrine changes**

- **Menopausal transition** – The menopausal transition (or perimenopause) begins, on average, four years before the final menstrual period (FMP) and is marked by irregular menstrual cycles, intense hormonal fluctuations, often accompanied by vasomotor complaints, sleep disturbances, and changes in sexual function. (See '[Menopausal transition \(perimenopause\)](#)' above.)

The early menopausal transition is characterized by a change in intermenstrual interval, an increase in serum follicle-stimulating hormone (FSH), and normal or high estradiol.

The late transition is characterized by more dramatic menstrual cycle changes and greater FSH and estradiol variability. Hot flashes are the most common symptom for women in the menopausal transition. They are most common in the late menopausal transition and early postmenopausal stage. Hot flashes are often associated with sleep disturbances. (See "[Menopausal hot flashes](#)".)

- **Menopause** – After the years of menstrual irregularity, women eventually experience permanent cessation of menses or clinical menopause. The FMP is defined retrospectively after 12 months of amenorrhea. (See '[Menopause](#)' above.)

- **Menopausal symptoms**

- **Hot flashes** – The most common symptom during the menopausal transition and menopause are hot flashes (also referred to as vasomotor symptoms or hot flushes), which occur in up to 80 percent of women. (See "[Menopausal hot flashes](#)" and '[Hot flashes](#)' above.)
- **Genitourinary atrophy symptoms** – Genitourinary atrophy symptoms, including vaginal dryness, dyspareunia, and sometimes sexual dysfunction, are most prevalent during the late menopausal transition and postmenopausal years. (See '[Genitourinary syndrome of menopause](#)' above and "[Overview of sexual dysfunction in females: Epidemiology, risk factors, and evaluation](#)".)
- **Mood symptoms** – Women during the menopausal transition appear to have a higher rate of mood symptoms than pre- or postmenopausal women. (See '[Depression](#)' above.)

- **Diagnosis of menopausal transition and menopause by patient age**

- **Staging system (STRAW)** – The Stages of Reproductive Aging Workshop (STRAW) staging system was developed based upon data from multiple longitudinal cohort studies. It is considered the gold standard for characterizing reproductive aging from the reproductive years through menopause and includes criteria for the reproductive years, the menopausal transition, FMP, and postmenopause based upon bleeding patterns, endocrine findings, and symptoms ([figure 2](#)). (See '[STRAW staging system](#)' above.)
- **Women over 45 years** – The diagnosis of the menopausal transition is made in women over 45 years based upon irregular menstrual cycles and menopausal symptoms such as hot flashes, mood changes, or sleep disturbance. We suggest no further diagnostic evaluation. Although serum FSH is often measured, it offers no additional information and may be misleading. (See '[Women over age 45 years](#)' above.)

Menopause may be diagnosed clinically as 12 months of amenorrhea in a woman over age 45 years in the absence of other biologic causes ([algorithm 1](#)). We do not recommend further diagnostic evaluation for women in this group. However, if there are any suggestive features of hyperprolactinemia or thyroid disease (galactorrhea, goiter, tachycardia, proptosis, etc), additional endocrine testing (eg, prolactin and thyroid-stimulating hormone [TSH]) should be performed. (See '[Diagnosis](#)' above.)

- **Women ages 40 to ≤45 years** – Women between the ages of 40 and 45 years who present with irregular menstrual cycles and menopausal symptoms may be experiencing early menopause or an earlier than anticipated menopausal transition. However, for women in this age group, with or without menopausal symptoms, we recommend the same endocrine evaluation as for any woman with oligo/amenorrhea: serum human chorionic gonadotropin (hCG), prolactin, TSH, and FSH ([algorithm 1](#)). (See '[Early menopause: Ages 40 to 45 years](#)' above.)
- **Women under age 40 years** – For women under age 40 years with irregular menses and menopausal symptoms, we recommend a complete evaluation for primary ovarian insufficiency. (See "[Clinical manifestations and diagnosis of primary ovarian insufficiency \(premature ovarian failure\)](#)" and '[Primary ovarian insufficiency: Age <40 years](#)' above.)

Use of UpToDate is subject to the [Terms of Use](#).

REFERENCES

1. Harlow SD, Gass M, Hall JE, et al. Executive summary of the Stages of Reproductive Aging Workshop + 10: addressing the unfinished agenda of staging reproductive aging. *J Clin Endocrinol Metab* 2012; 97:1159.
2. Bromberger JT, Schott LL, Kravitz HM, et al. Longitudinal change in reproductive hormones and depressive symptoms across the menopausal transition: results from the

- Study of Women's Health Across the Nation (SWAN). Arch Gen Psychiatry 2010; 67:598.
3. Freeman EW, Grisso JA, Berlin J, et al. Symptom reports from a cohort of African American and white women in the late reproductive years. Menopause 2001; 8:33.
 4. Hollander LE, Freeman EW, Sammel MD, et al. Sleep quality, estradiol levels, and behavioral factors in late reproductive age women. Obstet Gynecol 2001; 98:391.
 5. Lee CG, Carr MC, Murdoch SJ, et al. Adipokines, inflammation, and visceral adiposity across the menopausal transition: a prospective study. J Clin Endocrinol Metab 2009; 94:1104.
 6. Maki PM, Freeman EW, Greendale GA, et al. Summary of the National Institute on Aging-sponsored conference on depressive symptoms and cognitive complaints in the menopausal transition. Menopause 2010; 17:815.
 7. Santoro N, Brockwell S, Johnston J, et al. Helping midlife women predict the onset of the final menses: SWAN, the Study of Women's Health Across the Nation. Menopause 2007; 14:415.
 8. Matthews KA, Crawford SL, Chae CU, et al. Are changes in cardiovascular disease risk factors in midlife women due to chronological aging or to the menopausal transition? J Am Coll Cardiol 2009; 54:2366.
 9. Neer RM, SWAN Investigators. Bone loss across the menopausal transition. Ann N Y Acad Sci 2010; 1192:66.
 10. Woods NF, Mitchell ES. Sleep symptoms during the menopausal transition and early postmenopause: observations from the Seattle Midlife Women's Health Study. Sleep 2010; 33:539.
 11. Randolph JF Jr, Zheng H, Sowers MR, et al. Change in follicle-stimulating hormone and estradiol across the menopausal transition: effect of age at the final menstrual period. J Clin Endocrinol Metab 2011; 96:746.
 12. Randolph JF Jr, Sowers M, Gold EB, et al. Reproductive hormones in the early menopausal transition: relationship to ethnicity, body size, and menopausal status. J Clin Endocrinol Metab 2003; 88:1516.
 13. Manson JM, Sammel MD, Freeman EW, Grisso JA. Racial differences in sex hormone levels in women approaching the transition to menopause. Fertil Steril 2001; 75:297.
 14. Su HI, Sammel MD, Freeman EW, et al. Body size affects measures of ovarian reserve in late reproductive age women. Menopause 2008; 15:857.
 15. Van Voorhis BJ, Santoro N, Harlow S, et al. The relationship of bleeding patterns to daily reproductive hormones in women approaching menopause. Obstet Gynecol 2008; 112:101.
 16. Gold EB, Colvin A, Avis N, et al. Longitudinal analysis of the association between vasomotor symptoms and race/ethnicity across the menopausal transition: study of

- women's health across the nation. *Am J Public Health* 2006; 96:1226.
17. Randolph JF Jr, Sowers M, Bondarenko I, et al. The relationship of longitudinal change in reproductive hormones and vasomotor symptoms during the menopausal transition. *J Clin Endocrinol Metab* 2005; 90:6106.
 18. Thurston RC, Joffe H. Vasomotor symptoms and menopause: findings from the Study of Women's Health across the Nation. *Obstet Gynecol Clin North Am* 2011; 38:489.
 19. Bromberger JT, Assmann SF, Avis NE, et al. Persistent mood symptoms in a multiethnic community cohort of pre- and perimenopausal women. *Am J Epidemiol* 2003; 158:347.
 20. Greendale GA, Derby CA, Maki PM. Perimenopause and cognition. *Obstet Gynecol Clin North Am* 2011; 38:519.
 21. Derby CA, Crawford SL, Pasternak RC, et al. Lipid changes during the menopause transition in relation to age and weight: the Study of Women's Health Across the Nation. *Am J Epidemiol* 2009; 169:1352.
 22. Woodard GA, Brooks MM, Barinas-Mitchell E, et al. Lipids, menopause, and early atherosclerosis in Study of Women's Health Across the Nation Heart women. *Menopause* 2011; 18:376.
 23. Sternfeld B, Wang H, Quesenberry CP Jr, et al. Physical activity and changes in weight and waist circumference in midlife women: findings from the Study of Women's Health Across the Nation. *Am J Epidemiol* 2004; 160:912.
 24. Randolph JF Jr, Crawford S, Dennerstein L, et al. The value of follicle-stimulating hormone concentration and clinical findings as markers of the late menopausal transition. *J Clin Endocrinol Metab* 2006; 91:3034.
 25. Welt CK, McNicholl DJ, Taylor AE, Hall JE. Female reproductive aging is marked by decreased secretion of dimeric inhibin. *J Clin Endocrinol Metab* 1999; 84:105.
 26. Sherman BM, Korenman SG. Hormonal characteristics of the human menstrual cycle throughout reproductive life. *J Clin Invest* 1975; 55:699.
 27. Taffe JR, Dennerstein L. Menstrual patterns leading to the final menstrual period. *Menopause* 2002; 9:32.
 28. Miro F, Parker SW, Aspinall LJ, et al. Origins and consequences of the elongation of the human menstrual cycle during the menopausal transition: the FREEDOM Study. *J Clin Endocrinol Metab* 2004; 89:4910.
 29. Freeman EW, Sammel MD, Gracia CR, et al. Follicular phase hormone levels and menstrual bleeding status in the approach to menopause. *Fertil Steril* 2005; 83:383.
 30. Burger HG, Hale GE, Dennerstein L, Robertson DM. Cycle and hormone changes during perimenopause: the key role of ovarian function. *Menopause* 2008; 15:603.
 31. Burger HG. Unpredictable endocrinology of the menopause transition: clinical, diagnostic

- and management implications. *Menopause Int* 2011; 17:153.
32. McKinlay SM, Brambilla DJ, Posner JG. The normal menopause transition. *Maturitas* 1992; 14:103.
 33. Hee J, MacNaughton J, Bangah M, Burger HG. Perimenopausal patterns of gonadotrophins, immunoreactive inhibin, oestradiol and progesterone. *Maturitas* 1993; 18:9.
 34. Hale GE, Manconi F, Luscombe G, Fraser IS. Quantitative measurements of menstrual blood loss in ovulatory and anovulatory cycles in middle- and late-reproductive age and the menopausal transition. *Obstet Gynecol* 2010; 115:249.
 35. van Disseldorp J, Faddy MJ, Themmen AP, et al. Relationship of serum antimüllerian hormone concentration to age at menopause. *J Clin Endocrinol Metab* 2008; 93:2129.
 36. Hall JE. Neuroendocrine physiology of the early and late menopause. *Endocrinol Metab Clin North Am* 2004; 33:637.
 37. Greendale GA, Ishii S, Huang MH, Karlamangla AS. Predicting the timeline to the final menstrual period: the study of women's health across the nation. *J Clin Endocrinol Metab* 2013; 98:1483.
 38. Broer SL, Eijkemans MJ, Scheffer GJ, et al. Anti-müllerian hormone predicts menopause: a long-term follow-up study in normoovulatory women. *J Clin Endocrinol Metab* 2011; 96:2532.
 39. La Marca A, Sighinolfi G, Papaleo E, et al. Prediction of age at menopause from assessment of ovarian reserve may be improved by using body mass index and smoking status. *PLoS One* 2013; 8:e57005.
 40. Finkelstein JS, Lee H, Karlamangla A, et al. Antimüllerian Hormone and Impending Menopause in Late Reproductive Age: The Study of Women's Health Across the Nation. *J Clin Endocrinol Metab* 2020; 105:e1862.
 41. Woods NF, Mitchell ES. Symptoms during the perimenopause: prevalence, severity, trajectory, and significance in women's lives. *Am J Med* 2005; 118 Suppl 12B:14.
 42. National Institutes of Health. National Institutes of Health State-of-the-Science Conference statement: management of menopause-related symptoms. *Ann Intern Med* 2005; 142:1003.
 43. Dennerstein L, Dudley EC, Hopper JL, et al. A prospective population-based study of menopausal symptoms. *Obstet Gynecol* 2000; 96:351.
 44. Soules MR, Sherman S, Parrott E, et al. Executive summary: Stages of Reproductive Aging Workshop (STRAW). *Fertil Steril* 2001; 76:874.
 45. Kronenberg F. Hot flashes: epidemiology and physiology. *Ann N Y Acad Sci* 1990; 592:52.
 46. McKinlay SM. The normal menopause transition: an overview. *Maturitas* 1996; 23:137.

47. Nappi RE, Kroll R, Siddiqui E, et al. Global cross-sectional survey of women with vasomotor symptoms associated with menopause: prevalence and quality of life burden. *Menopause* 2021; 28:875.
48. Nappi RE, Siddiqui E, Todorova L, et al. Prevalence and quality-of-life burden of vasomotor symptoms associated with menopause: A European cross-sectional survey. *Maturitas* 2023; 167:66.
49. Avis NE, Crawford SL, Greendale G, et al. Duration of menopausal vasomotor symptoms over the menopause transition. *JAMA Intern Med* 2015; 175:531.
50. Williams RE, Levine KB, Kalilani L, et al. Menopause-specific questionnaire assessment in US population-based study shows negative impact on health-related quality of life. *Maturitas* 2009; 62:153.
51. Faubion SS, Enders F, Hedges MS, et al. Impact of Menopause Symptoms on Women in the Workplace. *Mayo Clin Proc* 2023; 98:833.
52. Bromberger JT, Meyer PM, Kravitz HM, et al. Psychologic distress and natural menopause: a multiethnic community study. *Am J Public Health* 2001; 91:1435.
53. Avis NE, Brambilla D, McKinlay SM, Vass K. A longitudinal analysis of the association between menopause and depression. Results from the Massachusetts Women's Health Study. *Ann Epidemiol* 1994; 4:214.
54. Schmidt PJ, Haq N, Rubinow DR. A longitudinal evaluation of the relationship between reproductive status and mood in perimenopausal women. *Am J Psychiatry* 2004; 161:2238.
55. Cohen LS, Soares CN, Vitonis AF, et al. Risk for new onset of depression during the menopausal transition: the Harvard study of moods and cycles. *Arch Gen Psychiatry* 2006; 63:385.
56. Freeman EW, Sammel MD, Lin H, Nelson DB. Associations of hormones and menopausal status with depressed mood in women with no history of depression. *Arch Gen Psychiatry* 2006; 63:375.
57. Freeman EW, Sammel MD, Liu L, et al. Hormones and menopausal status as predictors of depression in women in transition to menopause. *Arch Gen Psychiatry* 2004; 61:62.
58. Cohen LS, Soares CN, Joffe H. Diagnosis and management of mood disorders during the menopausal transition. *Am J Med* 2005; 118 Suppl 12B:93.
59. Hansen KR, Knowlton NS, Thyer AC, et al. A new model of reproductive aging: the decline in ovarian non-growing follicle number from birth to menopause. *Hum Reprod* 2008; 23:699.
60. Kravitz HM, Ganz PA, Bromberger J, et al. Sleep difficulty in women at midlife: a community survey of sleep and the menopausal transition. *Menopause* 2003; 10:19.
61. Freedman RR, Roehrs TA. Sleep disturbance in menopause. *Menopause* 2007; 14:826.

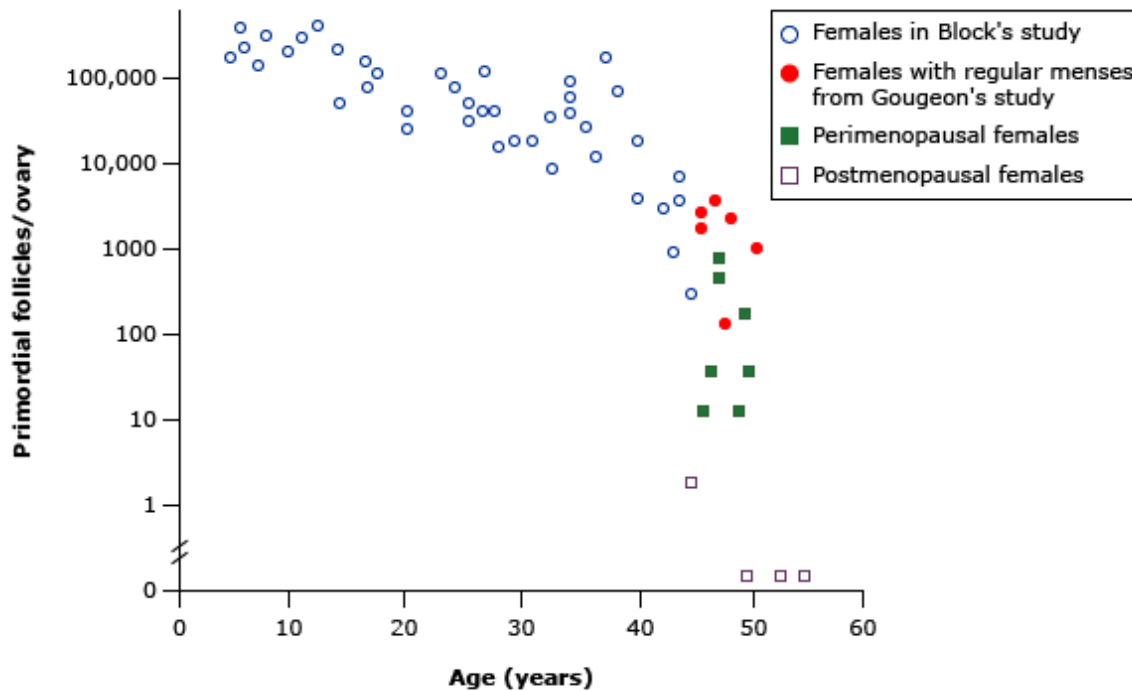
62. Juang KD, Wang SJ, Lu SR, et al. Hot flashes are associated with psychological symptoms of anxiety and depression in peri- and post- but not premenopausal women. *Maturitas* 2005; 52:119.
63. Joffe H, Hall JE, Soares CN, et al. Vasomotor symptoms are associated with depression in perimenopausal women seeking primary care. *Menopause* 2002; 9:392.
64. Greendale GA, Karlamangla AS, Maki PM. The Menopause Transition and Cognition. *JAMA* 2020; 323:1495.
65. Rentz DM, Weiss BK, Jacobs EG, et al. Sex differences in episodic memory in early midlife: impact of reproductive aging. *Menopause* 2017; 24:400.
66. Simon JA, Goldstein I, Kim NN, et al. The role of androgens in the treatment of genitourinary syndrome of menopause (GSM): International Society for the Study of Women's Sexual Health (ISSWSH) expert consensus panel review. *Menopause* 2018; 25:837.
67. Palacios S, Combalia J, Emsellem C, et al. Therapies for the management of genitourinary syndrome of menopause. *Post Reprod Health* 2020; 26:32.
68. Traish AM, Vignozzi L, Simon JA, et al. Role of Androgens in Female Genitourinary Tissue Structure and Function: Implications in the Genitourinary Syndrome of Menopause. *Sex Med Rev* 2018; 6:558.
69. Szoek CE, Cicuttini F, Guthrie J, Dennerstein L. Self-reported arthritis and the menopause. *Climacteric* 2005; 8:49.
70. Matthews KA, Wing RR, Kuller LH, et al. Influence of the perimenopause on cardiovascular risk factors and symptoms of middle-aged healthy women. *Arch Intern Med* 1994; 154:2349.
71. Blümel JE, Chedraui P, Baron G, et al. Menopause could be involved in the pathogenesis of muscle and joint aches in mid-aged women. *Maturitas* 2013; 75:94.
72. Szoek CE, Cicuttini FM, Guthrie JR, Dennerstein L. The relationship of reports of aches and joint pains to the menopausal transition: a longitudinal study. *Climacteric* 2008; 11:55.
73. Dugan SA, Powell LH, Kravitz HM, et al. Musculoskeletal pain and menopausal status. *Clin J Pain* 2006; 22:325.
74. Barnabei VM, Cochrane BB, Aragaki AK, et al. Menopausal symptoms and treatment-related effects of estrogen and progestin in the Women's Health Initiative. *Obstet Gynecol* 2005; 105:1063.
75. Chlebowski RT, Cirillo DJ, Eaton CB, et al. Estrogen alone and joint symptoms in the Women's Health Initiative randomized trial. *Menopause* 2013; 20:600.
76. MacGregor EA. Menstruation, sex hormones, and migraine. *Neurol Clin* 1997; 15:125.

77. Elting MW, Korsen TJ, Rekers-Mombarg LT, Schoemaker J. Women with polycystic ovary syndrome gain regular menstrual cycles when ageing. *Hum Reprod* 2000; 15:24.
78. Elting MW, Kwee J, Korsen TJ, et al. Aging women with polycystic ovary syndrome who achieve regular menstrual cycles have a smaller follicle cohort than those who continue to have irregular cycles. *Fertil Steril* 2003; 79:1154.
79. Black A, Francoeur D, Rowe T, et al. SOGC clinical practice guidelines: Canadian contraception consensus. *J Obstet Gynaecol Can* 2004; 26:219.
80. Mohyi D, Tabassi K, Simon J. Differential diagnosis of hot flashes. *Maturitas* 1997; 27:203.

Topic 7395 Version 38.0

GRAPHICS

Figure 1: Declining follicle number with age



A comparison of the relationship between age and primordial follicle number in Block's study of 44 females aged 7 to 44 years with that of Gougeon's study of females aged 45 to 55 years. Follicle depletion appears to accelerate in the decade preceding menopause.

Data from:

1. Block E. Quantitative morphological investigations of the follicular system in women; variations at different ages. *Acta Anat* 1952; 14:108.
2. Gougeon A. Caractères qualitatifs et quantitatifs de la population folliculaire dans l'ovaire humain adulte. *Contr Fert Sex* 1984; 12:527.

Figure 2: The Stages of Reproductive Aging Workshop +10 staging system for reproductive aging in females

Menarche					FMP (0)					
Stage	-5	-4	-3b	-3a	-2	-1	+1a	+1b	+1c	+2
Terminology	REPRODUCTIVE				MENOPAUSAL TRANSITION		POSTMENOPAUSE			
	Early	Peak	Late		Early	Late	Early			Late
					Perimenopause					
Duration	Variable				Variable	1-3 years	2 years (1+1)	3-6 years	Remaining lifespan	
PRINCIPAL CRITERIA										
Menstrual cycle	Variable to regular	Regular	Regular	Subtle changes in flow/ strength	Variable length: Persistent ≥7-day difference in length of consecutive cycles	Interval of amenorrhea of ≥60 days				
SUPPORTIVE CRITERIA										
Endocrine FSH AMH Inhibin B			Low Low	Variable* Low Low	↑ Variable* Low Low	↑ >25 international units/L ¶ Low Low	↑ Variable Low Low	Stabilizes Very low Very low		
Antral follicle count			Low	Low	Low	Low	Very low	Very low		
DESCRIPTIVE CHARACTERISTICS										
Symptoms						Vasomotor symptoms likely	Vasomotor symptoms most likely		Increasing symptoms of urogenital atrophy	

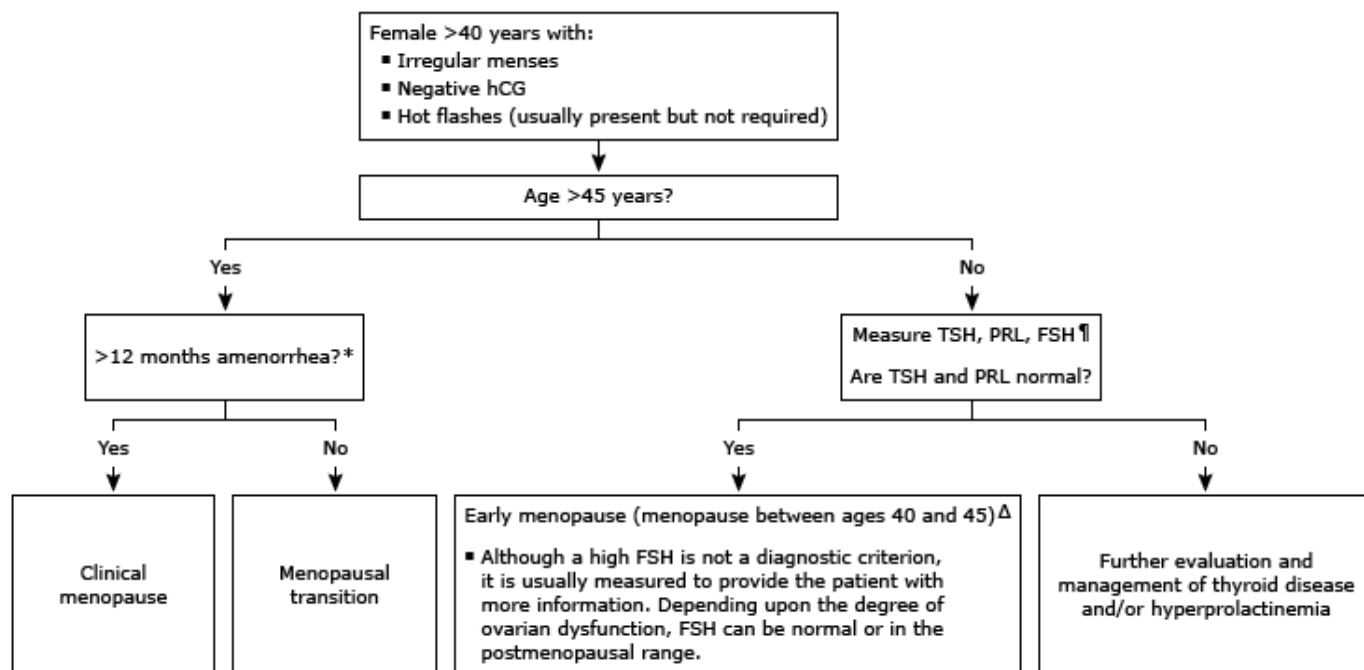
FMP: final menstrual period; FSH: follicle-stimulating hormone; AMH: anti-müllerian hormone; Arrow: elevated.

* Blood draw on cycle days 2 to 5.

¶ Approximate expected level based on assays using current international pituitary standard.

Reproduced with permission from: Harlow SD, Gass M, Hall JE, et al. Executive Summary of the Stages of Reproductive Aging Workshop + 10: Addressing the Unfinished Agenda of Staging Reproductive Aging. J Clin Endocrinol Metab 2012. Copyright © 2012 The Endocrine Society.

Algorithm 1: Evaluation of suspected menopause in females >40 years



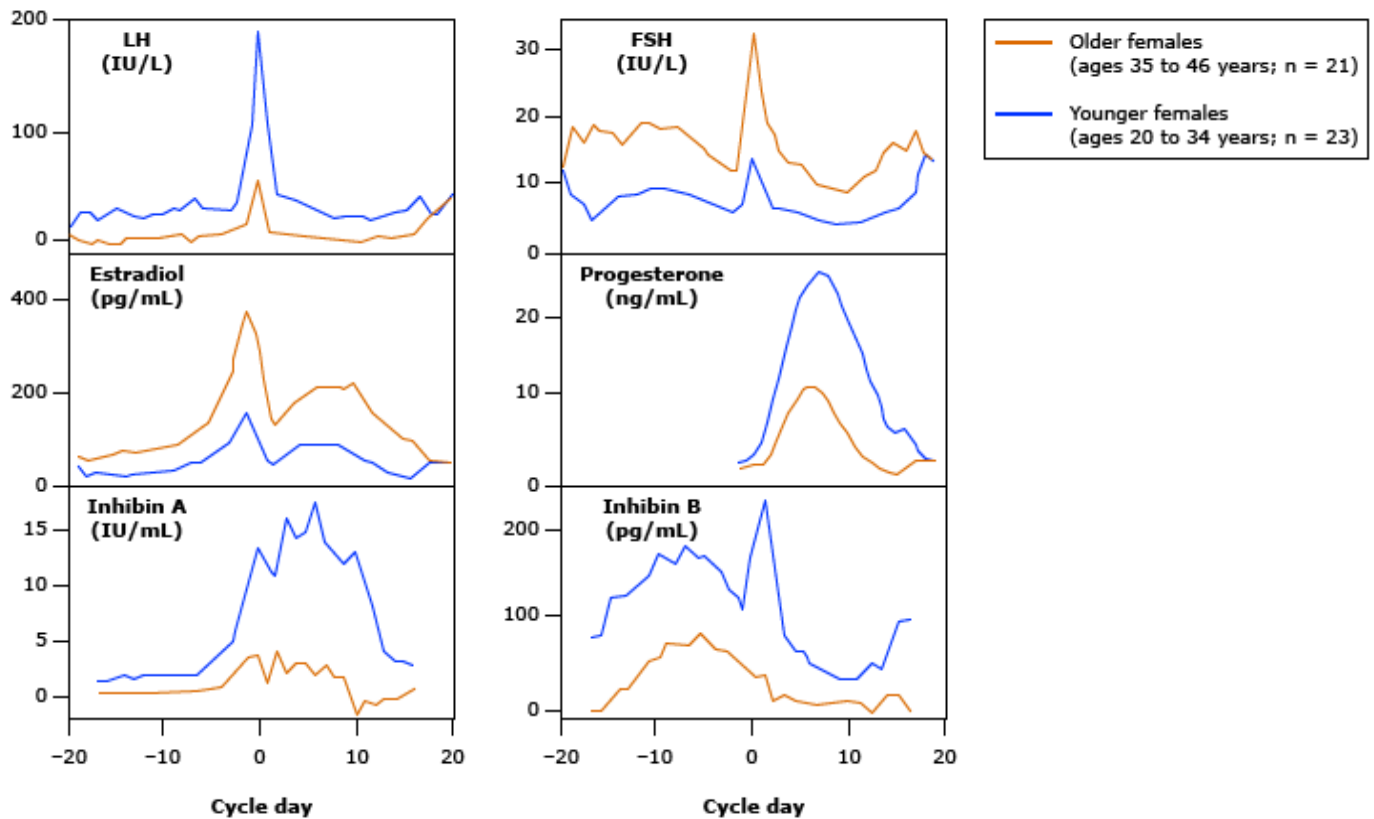
hCG: human chorionic gonadotropin; TSH: thyroid-stimulating hormone; PRL: prolactin; FSH: follicle-stimulating hormone.

* The diagnosis of perimenopause/menopause in females over age 45 years is clinical and based upon the presence of irregular menses and/or menopausal symptoms. Laboratory evaluation is not necessary or suggested.

¶ In females ages 40 to 45, the presence of irregular menses and hot flashes strongly suggests menopause. However, laboratory evaluation should be performed first to rule out other endocrine disorders.

Δ Diagnostic criteria for early menopause: irregular menses, hot flashes, without another endocrine disorder (thyroid disease, hyperprolactinemia).

Figure 3: Hormone levels: Older and younger reproductive age

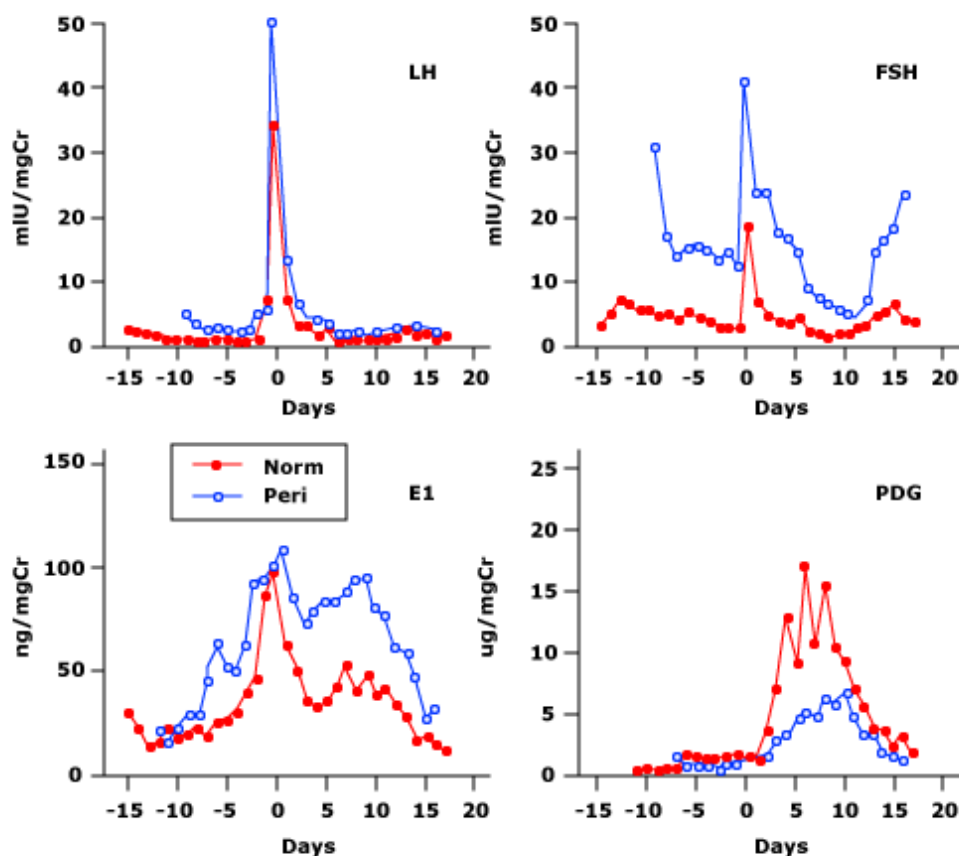


Mean daily levels of gonadotropins, sex steroids, and inhibins in older (ages 35 to 46 years; n = 21), shown in orange, and younger females (ages 20 to 34 years; n = 23), shown in blue.

FSH: follicle-stimulating hormone; LH: luteinizing hormone.

Adapted from: Welt CK, McNicholl DJ, Taylor AE, Hall JE. Female reproductive aging is marked by decreased secretion of dimeric inhibin. J Clin Endocrinol Metab 1999; 84:105.

Figure 4: Early transition hormone patterns

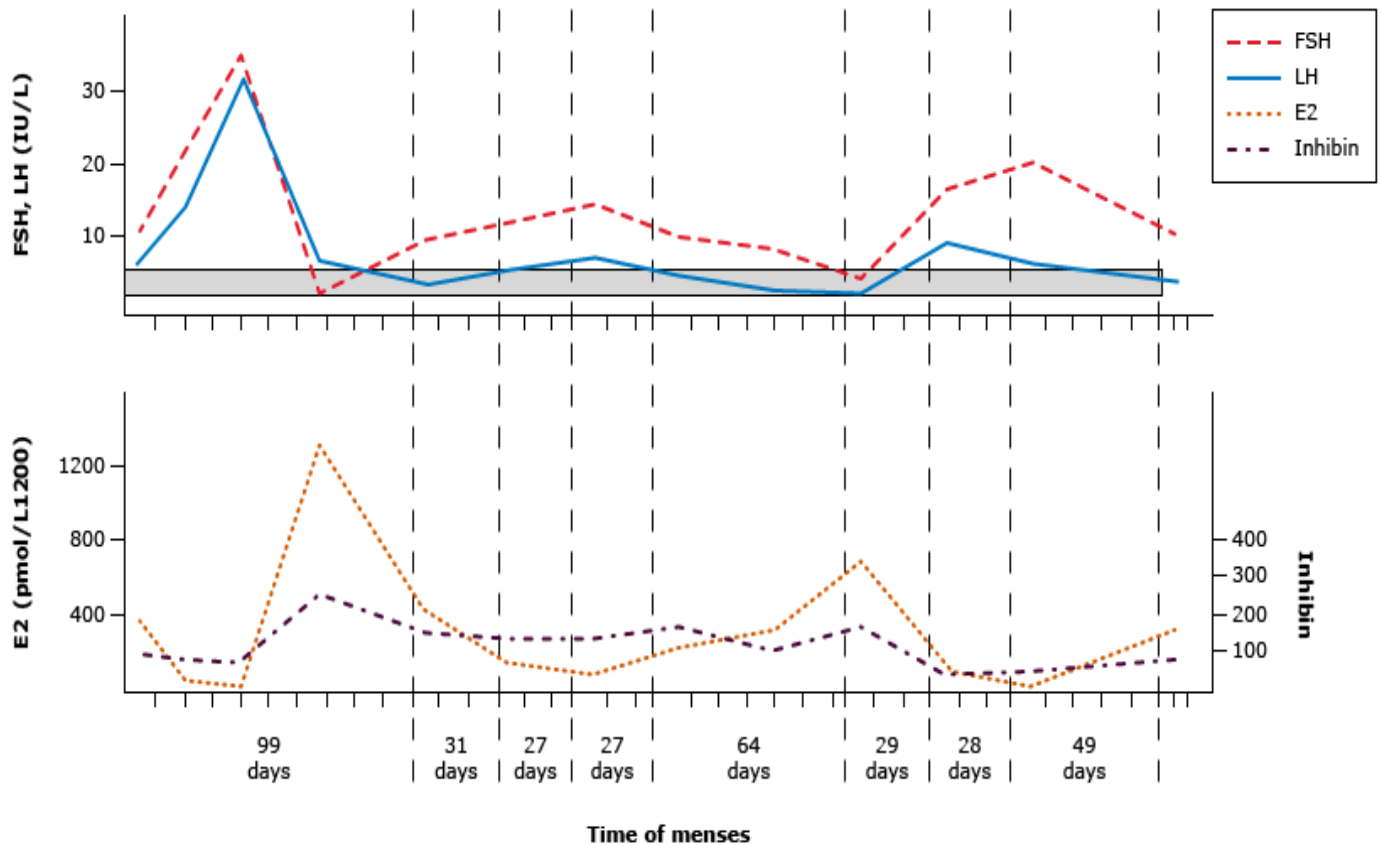


Urinary reproductive hormone patterns in a group of 11 mid-reproductive-age (norm, age 19 to 38 years) and 11 late reproductive-age (peri, age 43 to 52 years) females having ovulatory cycles. Note that in late reproductive-age females, FSH is consistently elevated throughout the cycle, estradiol metabolite excretion (E1c) is overall elevated, and progesterone metabolites (PDG) are decreased compared with the mid-reproductive-age female.

LH: luteinizing hormone; FSH: follicle-stimulating hormone; E1: estrone; PDG: pregnanediol glucuronide; E1c: estrone conjugate.

Reproduced with permission from: Santoro N, Brown JR, Adel T, Skurnick JH. Characterization of Reproductive Hormonal Dynamics in the Perimenopause. J Clin Endocrinol Metab 1996; 81:1495. Copyright © 1996 by The Endocrine Society.

Figure 5: Onset of the menopausal transition

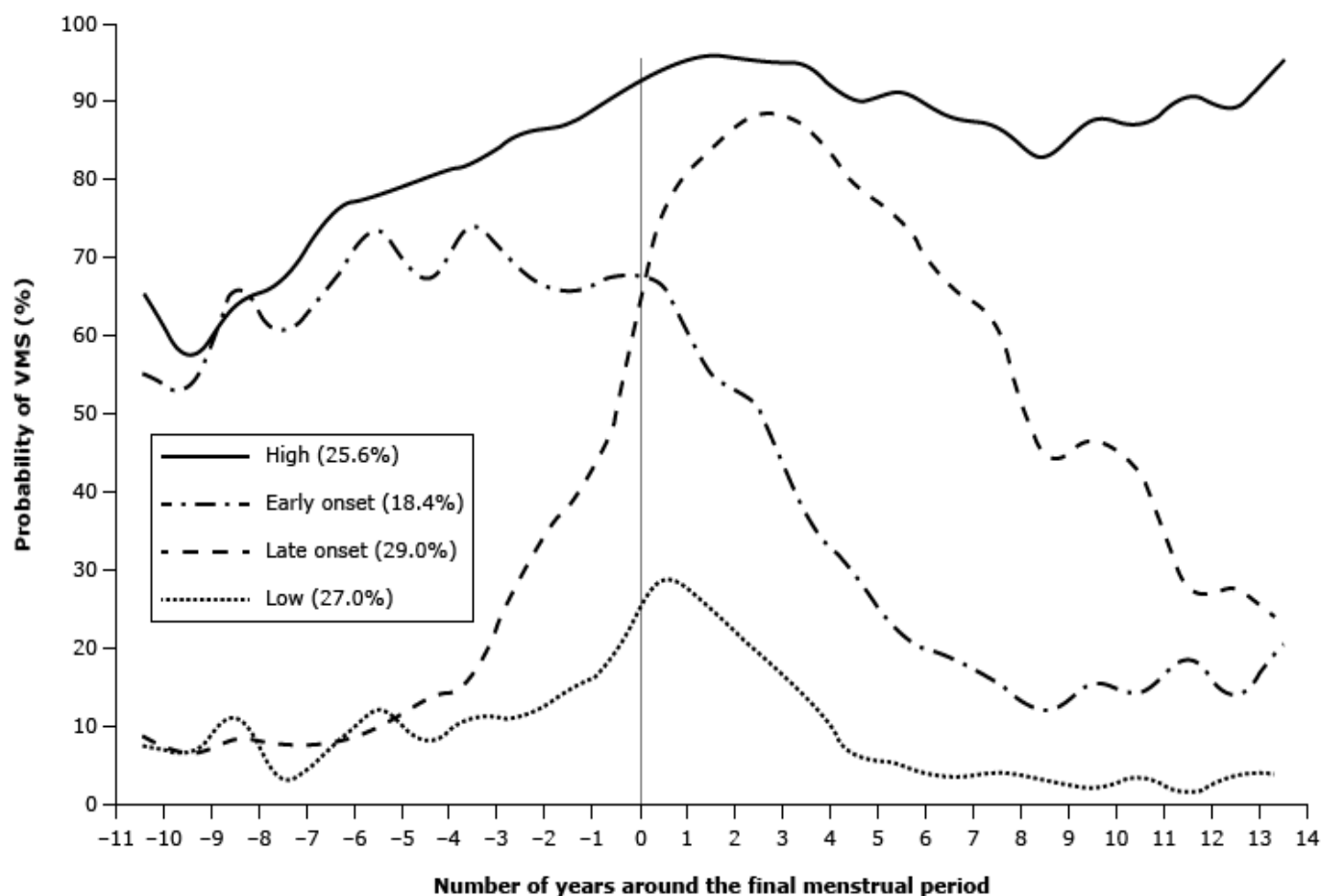


Concentrations of FSH, LH, E2, and inhibin in a volunteer studied at the onset of menopausal transition. The vertical dashed lines represent the times of menses. The horizontal shaded bar represents the young-normal range for FSH. Note the marked fluctuations in hormone levels.

E2: estradiol; FSH: follicle-stimulating hormone; LH: luteinizing hormone.

Reproduced with permission from: Hee J, MacNaughton J, Bangah M, Burger HG. Perimenopausal patterns of gonadotropins, immunoreactive inhibin, estradiol and progesterone, Maturitas 1993; 18:9. Copyright © 1993 Elsevier.

Figure 6: Trajectories of vasomotor symptoms over the menopause transition



Probability of VMS represents the average observed probability of VMS at each time point within each trajectory subgroup. No factors were included in the model.

VMS: vasomotor symptoms.

From: Tepper PG, Brooks MM, Randolph JF, et al. Characterizing the trajectories of vasomotor symptoms across the menopausal transition. *Menopause* 2016 23:1067. DOI: [10.1097/GME.0000000000000676](https://doi.org/10.1097/GME.0000000000000676). Copyright © 2016 The North American Menopause Society. Reproduced with permission from Lippincott Williams & Wilkins. Unauthorized reproduction of this material is prohibited.

Contributor Disclosures

Robert F Casper, MD Stocks: TRIO Fertility [Infertility]. Consultant/Advisory Boards: Abbott International [Infertility]; AbbVie [Endometriosis]; EMD-Serono [In vitro fertilization]; Ferring Pharmaceuticals [In vitro fertilization]. All of the relevant financial relationships listed have been mitigated. **Robert L Barbieri, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Jean E Mulder, MD** No relevant financial relationship(s) with ineligible companies to disclose.

Contributor disclosures are reviewed for conflicts of interest by the editorial group. When found, these are addressed by vetting through a multi-level review process, and through requirements for references to be provided to support the content. Appropriately referenced content is required of all authors and must conform to UpToDate standards of evidence.

[Conflict of interest policy](#)

→